CSC 350 Assignment 6
Spring 2025

Due: Tuesday, April 15 by 11:59 PM

General Instructions

- Scan or compile snapshots of your submission into a PDF and upload it to Canvas.
- Assignments are **due 11:59 PM** on the scheduled due date; generally, late assignments will not be accepted.
- Unless specified, **justify your answers always**. If justifications are missing or do not show reasonably how your answers are derived, 0 points will be assigned regardless of the correctness of the answers.
- Write **neatly** in the event you write your answers on paper. **Especially, for table completion questions, if your answers are not given neatly in well-drawn tables, the answers won't be graded**.
- **This assignment must be done by yourself without any unauthorized assistance**. Write your full name and numeric university ID on the front page of your submission, along with the following pledge statement.

I pledge my honor that all parts of this assignment were done by me individually and without any unauthorized assistance or collaboration with others.

Q1. Verify that each of the following relations is an equivalence relation or find the properties that are not satisfied.

(a) $s \sim t$ iff s and t are strings with the same occurrences of each letter.

(b) $a \; R \; b$ iff either $a \, mod \, 4 = b \, mod \, 4$ or $a \, mod \, 6 = b \, mod \, 6$, over **N**.

Q2. Describe the equivalence classes for the following relation on **N**.

$x \sim y$ iff $x \, mod \, 2 = y \, mod \, 2$ and $x \, mod \, 6 = y \, mod \, 6$.

Q3. For each of the following relations
1: State whether it is a partial order or not
2: Give the reason for your answer

(a) isParentOf

(b) isOlderThan

(c) isSisterOf

(d) $\{(2,1),(1,3),(2,3)\}$

Q4. Given the following poset with the subset relation,

$P = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}\}$

(a) Draw a poset diagram.

(b) Describe the `glb` and `lub` of pairs of elements in terms of set operations.
That is, given two elements S and T in the above poset, what is the `glb` of S and T?
What is the `lub` of S and T ?

Hint: $S \cup T$ is the `lub` of S and T.

Q5. Consider the poset $\langle S, R \rangle$, with $S = \{a,b,c,d,e,f,g,h\}$ and relation

$R = \{(a,e),(b,d),(b,f),(c,a),(d,c),(d,h),(f,c),(g,b),(h,a)\}$

(only the immediate predecessor / successor pairs are listed).

(a) Use the topological sort algorithm (of text) to obtain a sorted order of these elements that is consistent with the given partial order. Specify the initial values, and what happens for the first 3 nodes.

(b) Is the sequence (`gbfcdhae`) consistent with the given partial order? Why?